

EECS2311 Take-Home Assignment
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*****Note: The group only had implemented three user stories as a result we have assigned 2 people per user story. (verified with TA)**

User Story: Map of Photos

As a photographer, I want to be able to view a map of photos in different locations uploaded by others so I don't have to always scout for locations myself.

End-to-End Manual Test Cases

Test Case: TC-Map-01 - Load map with photo markers

****Steps:****

1. Launch the app
2. Ensure `photoCollection` contains valid lat/long data
3. Navigate to the map component

****Expected Result:****

Map loads centered and markers appear.

****Actual Result:****

Map loads centered and markers appear.

****Pass/Fail:****

Pass

Test Case: TC-Map-02 - Marker popup displays image and description

****Steps:****

1. Click on a visible photo marker
2. Observe the popup

****Expected Result:****

No popup is implemented. Clicking does nothing.

****Actual Result:****

No popup is implemented. Clicking does nothing.

****Pass/Fail:****

Fail

Test Case: TC-Map-03 - No photoCollection data

****Steps:****

1. Launch the app
2. Pass an empty array to the Map component

****Expected Result:****

Map loads without errors. No markers appear.

****Actual Result:****

Map loads without errors. No markers appear.

****Pass/Fail:****

Pass

Test Case: TC-Map-04 - Invalid coordinates are ignored or handled gracefully

****Steps:****

1. Launch the app
2. Pass photo with invalid lat: 999, long: 999

****Expected Result:****

Markers with invalid coordinates are rendered. No validation exists.

****Actual Result:****

Markers with invalid coordinates are rendered. No validation exists.

****Pass/Fail:****

Fail

Test Case: TC-Map-05 - Invalid icon URL used for marker

****Steps:****

1. Replace iconUrl with a bad link
2. Launch the app

****Expected Result:****

Marker loads with broken icon. No crash.

****Actual Result:****

Marker loads with broken icon. No crash.

****Pass/Fail:****

Pass

Test Case: TC-Map-06 - Zoom control works properly

****Steps:****

1. Click zoom in/out controls
2. Try scrolling with trackpad

****Expected Result:****

Zoom works. Scroll is disabled by config.

****Actual Result:****

Zoom works. Scroll is disabled by config.

****Pass/Fail:****

Pass

Test Case: TC-Map-07 - Display of clustered or overlapping markers

****Steps:****

1. Load 5+ photos close together
2. Observe marker behavior

****Expected Result:****

Markers slightly overlap but all clickable.

****Actual Result:****

Markers slightly overlap but all clickable.

****Pass/Fail:****

Pass

Test Case: TC-Map-08 - Verify map renders after authentication

****Steps:****

1. Sign in
2. Navigate to map

****Expected Result:****

Map renders successfully after auth.

****Actual Result:****

Map renders successfully after auth.

****Pass/Fail:****

Pass

Test Case: TC-Map-09 - Clicking on non-marker area does not crash app

****Steps:****

1. Click random areas of the map

****Expected Result:****

No crash or reaction. Works as expected.

****Actual Result:****

No crash or reaction. Works as expected.

****Pass/Fail:****

Pass

Bug Reports

1. Title: Missing popup for photo markers

Steps: Click on any marker on the map

Expected: A popup appears with image and description

Actual: No popup is shown

Severity: Medium

2. Title: Invalid coordinates not handled

Steps: Pass invalid coordinates to a photo (e.g., lat: 999)

Expected: Marker should not render or app should log warning

Actual: Marker is rendered with invalid location

Severity: Low

Code Review & Code Smells

The following code smells were checked:

- Long functions
- Duplicated code
- Hardcoded values
- Missing functionality
- Logic in JSX

(References: SourceMaking, Refactoring.Guru, PragmaticWays)

Detected Smells:

1. Hardcoded Values:

- Map center and icon size are hardcoded in Map.tsx.

- Recommendation: Move to constants file or make them configurable.

2. Missing Popup in Marker:

- Markers don't have <Popup> elements.
- Recommendation: Add <Popup> to show image and description on click.

3. Logic in JSX:

- The photoCollection mapping logic is embedded directly inside JSX.
- Recommendation: Refactor the mapping logic outside for clarity and reusability.